Traumatic bilateral optic nerve avulsion in a teen

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A 16-year-old boy presented with bilateral vision loss with severe facial and head injuries after a road traffic accident. Vision was no light perception in both eyes. Examination revealed a complete optic nerve avulsion on the right side and a complete optic nerve avulsion with an intact nerve sheath attachment on the left side, confirmed by B-scan ultrasonography and magnetic resonance imaging. Follow-up visits showed extensive posterior pole gliosis in both eyes. This report emphasizes the critical importance of understanding the severity of injury and the underlying pathomechanism of bilateral optic nerve avulsion, which is associated with a uniformly poor visual prognosis.

Case Report

A 16-year-old boy presented to the emergency department with severe midfacial trauma, including nasal and auricular hemorrhage, following a high-impact road traffic accident. The alleged mechanism of injury involved a head-on collision between two two-wheelers, resulting in the patient falling face-down onto the ground. The patient reported a loss of consciousness for approximately 10 minutes, with a concurrent episode of post-traumatic amnesia. He underwent an open reduction and internal fixation of facial bone fractures.

On presentation at our institution, he was alert, coherent, and well-oriented, with stable systemic parameters. Incomplete mechanical ptosis was noted in both eyes, with an intraocular pressure of 10 mm Hg in the right and 9 mm Hg in the left eye. The right eye anterior segment was within normal limits. The left eye cornea showed inferior exposure keratopathy. The pupils were dilated and fixed in both eyes. Initial assessment prior to the surgery for facial bone fractures had revealed bilateral dilated and fixed pupils. Light perception was not detected in either eye. The right eye fundus showed a recessed optic nerve head with deep cavitation and peripapillary hemorrhages, with a taut vitreous and hemorrhage (Figure 1A). The left eye fundus showed a similar posterior recession of the optic nerve with deep cavitation, peripapillary hemorrhages, and an inferior subhyaloid hemorrhage (Figure 1B). The fundus features were consistent with a bilateral complete optic nerve avulsion.

B-scan ultrasonography showed a widened optic nerve head shadow with hypolucency at the optic nerve junction with the globe (Figure 2). Computed tomography of the orbits showed bilateral frontal bone/orbital roof fractures, nasal bone fractures, pneumo-orbits, and fractures of the sphenoid sinus, greater wing, and lesser wing of the sphenoid. Magnetic resonance imaging (MRI) of the orbits demonstrated complete avulsion of the right optic nerve, and the left eye showed disinsertion of the optic nerve from the globe with an intact optic nerve sheath (Figure 3A-B). In addition, a basifrontal lobe hematoma, pneumocephalus, pneumo-orbits, and extraocular muscle edema were noted on both sides (Figure 3C-D). Visual evoked potential (VEP) revealed nonrecordable waveforms, indicating a disruption in anterior visual pathways. At the 2 months' follow-up, gliotic proliferation covered the optic nerve area in both eyes, with vision remaining as no perception of light.

Discussion

Optic nerve avulsions involve the disinsertion of nerve fibers at the disk margins, and are typically encountered after road traffic accidents or work-related injuries and, rarely, following self-mutilation.[1-2] Optic nerve avulsion can be partial (only the optic nerve is torn) or complete (both the extraocular muscles and the optic nerve are torn, causing total luxation of the ocular bulb).[3] Most cases reported in literature are unilateral, presenting with partial or complete disinsertion of optic nerve fibers.[4,5] Avulsion is secondary to combined torsional globe rotation and anterior subluxation following orbital compressive injuries.[6] They tend to be unilateral as the maximum decompressive force causes fractures to occur on one side. The point of disinsertion is frequently at the junction of the optic nerve with the globe, where the nerve fibers lack myelin and meningeal cover.

Bilateral optic nerve avulsions are extremely rare and reported in severe facial injuries following road traffic accidents.[7,8] Such cases are associated with midfacial compressive fractures with luxation of globes. Multiple fractures of facial and orbital bones lead to compression in the posterior orbit, reducing orbit volume, avulsion of extraocular muscles, and luxating the globe anteriorly. Optic nerve avulsion in such injuries can be noted at the optic canal level or the orbital level due to the tethering effect of the optic nerve at the attachment of the dura and periosteum of the optic canal.[9-10]

In the present case, although there was no globe luxation, optic nerve disinsertion occurred at the level of the posterior globe in both eyes. The right eye had complete disinsertion of both the optic nerve and sheath, whereas the left eye showed only disinsertion of the optic nerve with an intact nerve sheath attached to the globe, as noted on MRI. There was no difference in the fundus features

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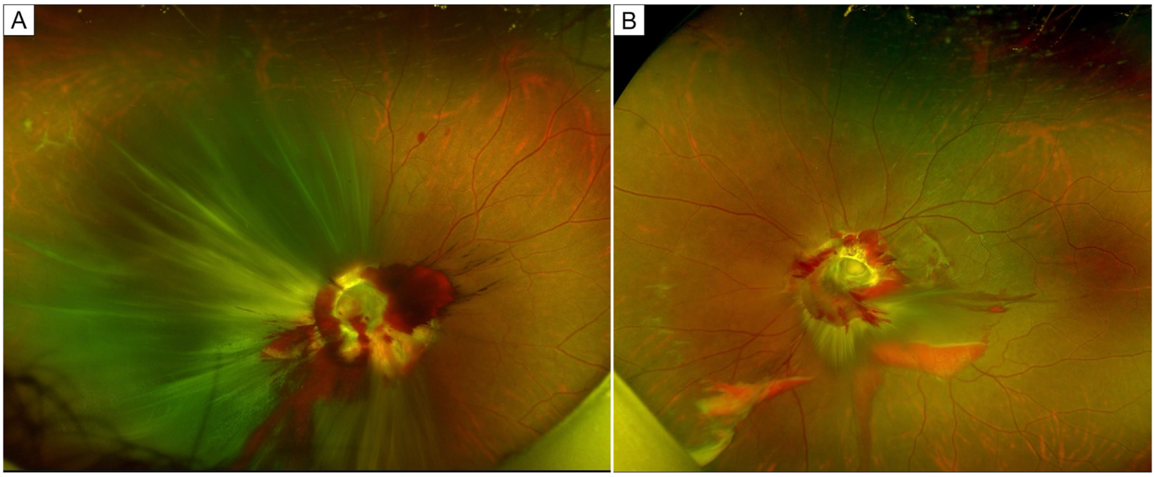

FIG 1. Optos images of the right eye (A) and left eye (B) showing deep excavation, with a recession of the optic nerve head and peripapillary and vitreous hemorrhage.

between the eyes with and without sheath avulsion, and vision remained no perception of light. The head-on collision, resulting in the patient's face-down fall, likely contributed to the optic nerve avulsion via a direct impact on the orbital roof and a shearing effect through a contrecoup mechanism at the optic canal. This proposed mechanism is consistent with static loading studies and blunt trauma simulations, which suggest that forehead impact generates a stress wave that propagates through the orbital roof and concentrates at the orbital apex, leading to increased susceptibility to optic nerve avulsion.[11]

Optic nerve avulsions carry a poor prognosis. In eyes with bilateral avulsion and globe luxation, the vascular supply to the globe is compromised, leading to phthisis bulbi over time. Gliosis sets in at the site of avulsion in bilateral optic nerve-avulsed eyes without globe subluxation. Successful treatment is not possible in patients with complete optic nerve avulsion. In contrast to full-thickness optic nerve avulsion, partial optic nerve avulsion is associated with a variable visual prognosis, as partial preservation of vision is possible depending on the extent of the optic nerve circumference involved. Patients may present with arcuate visual field defects due to associated vessel occlusion at the site of avulsion.[12,13]

This case report underscores the significance of anatomical and pathomechanical considerations in bilateral optic nerve avulsion. The proposed mechanism of injury, involving a direct impact on the orbital roof and a shearing effect through a contrecoup mechanism at the optic canal, is consistent with the observed patterns of optic nerve damage. The distinction between full-thickness and partial optic nerve avulsion is crucial, as the latter may exhibit a variable visual prognosis.

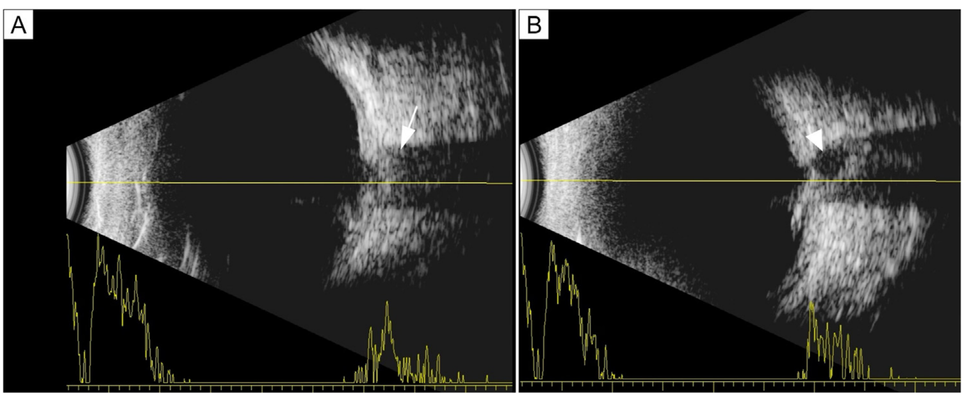

FIG 2. B-scan ultrasonography of the right eye (A) showing a widening of the optic nerve head shadow (arrow). The left eye (B) shows a widening and hypolucent area at the junction of the globe and nerve (arrowhead).

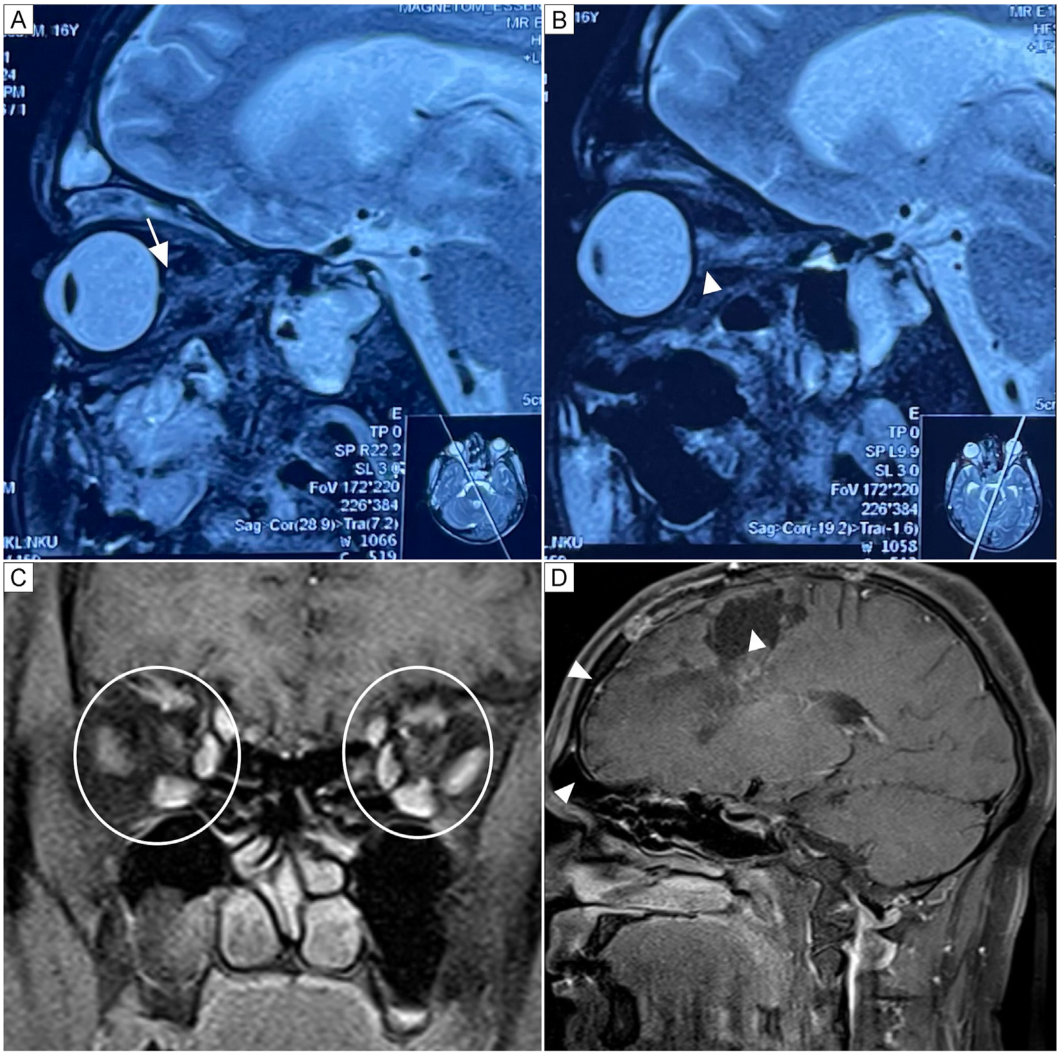

FIG 3. Sagittal plane magnetic resonance fat-saturated images of the right eye, central orbit portion (A), showing the complete avulsed optic nerve and the sheath at the retrobulbar portion (arrow). In the left eye (B) the sheath is attached to the posterior globe with avulsion of the optic nerve within the sheet (arrowhead). Coronal sections with contrast show extraocular muscle edema (C, circles), right basifrontal lobe hematoma, and pneumocephalus (D, arrows).

References

1. Foster BS, March GA, Lucarelli MJ, et al. Optic nerve avulsion. Arch Ophthalmol 1997;115:623-30.
2. Al Amry M, AlHijji L, Elkhamary SM, Mousa A, AlGaeed A, AlGhadeer H. Optic nerve avulsion: pattern and etiologies at a tertiary eye care center in Saudi Arabia: an 8-year retrospective study. Clin Ophthalmol 2023 Sep 6;17:2633-41.
3. Chia MA, Shah VH, Turner AW. Partial optic nerve avulsion: a diagnostic challenge. Clin Experiment Ophthalmol 2020;48:701-2.
4. Sindhu M, Bansal RK, Goyal P. Ultrasound findings in optic nerve avulsion. QJM 2020;113:363-4.
5. Simsek T, Simsek E, Ilhan B, Ozalp S, Sekercioglu B, Zilelioglu O. Traumatic optic nerve avulsion. J Pediatr Ophthalmol Strabismus 2006;43:367-9.
6. Tok L, Tok OY, Argun TC, et al. Bilateral traumatic globe luxation with optic nerve transection. Case Rep Ophthalmol 2014;5:429-34.
7. Lang GK, Bialasiewicz AA, Röhr WD. Bilateral traumatic eye avulsion [in German]. Klin Monbl Augenheilkd 1991;198:112-16.
8. Pujari A, Saxena R, Phuljhele S, Bhaskaran K, Basheer S, Sharma P. Pathomechanism of optic nerve avulsion. Med Hypotheses 2019; 125:28-30.
9. Pillai S, Mahmood MA, Limaye SR. Complete evulsion of the globe and optic nerve. Br J Ophthalmol 1987;71:69-72.
10. Şahin S, Furundaoturan O, Barış ME, Demirkılınç Biler E. Sheath-preserving complete optic nerve avulsion following closed-globe injury: a case report. Turk J Ophthalmol 2022;52:216-19.
11. Li Y, Singman E, McCulley T, Wu C, Daphalapurkar N. The biomechanics of indirect traumatic optic neuropathy using a computational head model with a biofidelic orbit. Front Neurol 2020;11:346.
12. Anandarajah HR, Seto S, Godiwalla RY, Chisholm SAM, Kim JE, Walsh RD. Acute and chronic optical coherence tomography findings in partial optic nerve head avulsion. J AAPOS 2024;104060.
13. Bayram-Suverza M, Rosano-Barragán M, Ramírez-Estudillo JA. Long-term follow-up of a patient with partial optic nerve avulsion associated with submacular hemorrhage who underwent pneumatic displacement. Am J Ophthalmol Case Rep 2024;35:102083.